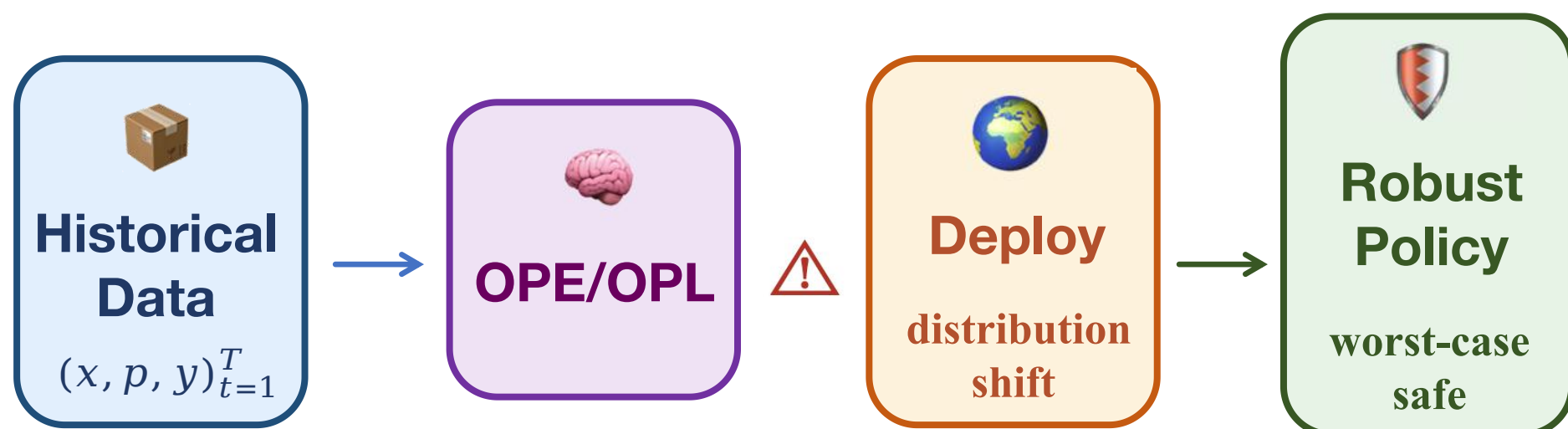


1. Introduction



X
Prior DRO (Discrete)
Inapplicable to continuous prices

X
DRO-IPW (Leung et al. 2025)
Sensitive to propensity estimation; slow regret

✓
Ours - OPE
Only 3 regressions, semiparameter efficient

✓
Ours - OPL
Sharper regret, slow nuisance estimation rates

Algorithm	Task	Key Guarantee
LDR²O²PE-CP	Evaluation	$\sqrt{Th}$ -consistent & semipar. efficient
CDR²O²PL-CP	Learning	Regret $\tilde{O}_p(T^{-s/(2s+1)})$, $s = 1, 2$

PRICING MODEL

Valuation: $v(X, Z) = f(X) + Z$

Purchase: $Y = 1[Z \geq P - f(X)]$ Revenue: $R = P \cdot Y$

Robust objective: $R_\delta(\pi) \triangleq \inf_{\mathbf{P} \in \mathcal{U}_{\mathbf{P}_0}(\delta)} \mathbb{E}_{\mathbf{P}}[R(\pi)]$

$R(\pi) \triangleq \mathbb{E}_{\mathbf{P}_0}[r(X, \pi(X), Z)]$

KL uncertainty set: $\mathcal{U}(\delta) = \{\mathbf{P} : D_{\text{KL}}(\mathbf{P} \parallel \mathbf{P}_0) \leq \delta\}$

DOUBLY ROBUST ESTIMATOR

$$\widehat{W}_T(\pi, \alpha) = \widehat{m}_0(x, \pi(x); \alpha) + \frac{K((p - \pi(x))/h)}{h} \widehat{\pi}_0(p|x) \cdot (e^{-py/\alpha} - \widehat{m}_0(x, p; \alpha))$$

2. Method

KEY INSIGHT

Discontinuous revenue $R = P \cdot 1[\cdot]$, but outcome regression $m_j(x, p; \alpha) = \mathbb{E}[(PY)^j \exp(-PY/\alpha) | X = x, P = p]$ is **smooth** via s -differentiability of noise CDF F_Z . Kernel smoothing + localization unlocks near-optimal rates.

DUAL REFORMULATION (STRONG DUALITY)

$R_\delta(\pi) = \sup_{\alpha \geq 0} -\alpha \log W(\pi, \alpha) - \alpha \delta$
 $W(\pi, \alpha) = \mathbb{E}[\exp(-r(X, \pi(X), Z)/\alpha)]$.
Reduce infinite-dim problem to scalar α .

LDR2O2PE-CP — Evaluation (3 REGRESSIONS ONLY)

1 Cross-fit Propensity $\widehat{\pi}_0(p|x)$

2 Localize at $\widehat{\alpha}_{\text{init}}$
Initial IPW estimate $\rightarrow$ fit $\widehat{m}_j$ once at this single point

3 Solve DR Moment Eq. (Newton-Raphson)
 $\rightarrow$ output $\widehat{\alpha}, \widehat{R}_\delta$

CDR2O2PL-CP — Learning

1 Continuum Regression via GRF Weights
 $\widehat{m}_0 = \sum_t \widehat{w}_t(x, p) \exp(-p_t y_t / \alpha)$

2 Alternating Optimization
Update $\widehat{\alpha} \leftarrow \arg\max_{\alpha \geq 0} -\alpha \log \widehat{W}_T(\widehat{\pi}, \alpha) - \alpha \delta$
then $\widehat{\pi} \leftarrow \arg\min_{\pi \in \Pi} \log \widehat{W}_T(\pi, \widehat{\alpha})$

3. Main Results

OPE CONVERGENCE

$\sqrt{Th}$ -consistent
Neyman Orthogonality
+ semipar. efficient

OPL REGRET

$\tilde{O}_p(T^{-s/(2s+1)})$
 $h = \Theta(T^{-1/(2s+1)})$

DRO regret improvement $\tilde{O}_p(T^{-1/4}) \rightarrow \tilde{O}_p(T^{-2/5})$ ($s=2$)

Method	Action	Robust	Regret
Kitagawa & Athey'18–21	Binary	X	$O_p(T^{-1/2})$
Dudik, Swaminathan...	Discrete	X	$O_p(T^{-1/2})$
Kallus, Colangelo, Ai...	Cts.	X	$O_p(T^{-s/(2s+1)})$
Kallus, Sinian...	Discrete	✓	$O_p(T^{-1/2})$
Leung'25 (DRO-IPW)	Cts.	✓	$O_p(T^{-1/4})$
Ours (DRO-DR)	Cts.	✓	$\tilde{O}_p(T^{-s/(2s+1)})$

THM 1 — OPE ASYMPTOTIC NORMALITY

Under product rate condition $\varrho_\pi + \varrho_\alpha \wedge \varrho_m \geq (1 - \gamma_h)/2$, and $h = o(T^{-1/(1+2s)})$:

$$\sqrt{Th}(\widehat{\theta} - \theta^*) \xrightarrow{d} N(0, \Sigma^*)$$

Σ^* achieves semiparametric efficiency lower bound. Weaker than the 1/2 threshold for discrete settings.

THM 2 — OPL REGRET (W.P. $\geq 1 - 7\epsilon$)

$$\text{Reg}(\widehat{\pi}) \lesssim \tilde{O}\left(\log(1/h) \cdot (d_\Pi/Th + \sqrt{d_\Pi/Th}) + T^{-\gamma_\pi - \gamma_m} + h^s\right)$$

Requires only $\gamma_\pi + \gamma_m \geq s/(2s+1)$, also improves the regret of Leung'25 IPW estimator.

4. Experiments

SIMULATION SETUP

CONTEXT: $X \in \mathbb{R}^3 \sim N(0, I_3)$

LOG. POLICY: $P|X \sim N(\mu_0(X), 1)$, $P \in [0.5, 3]$

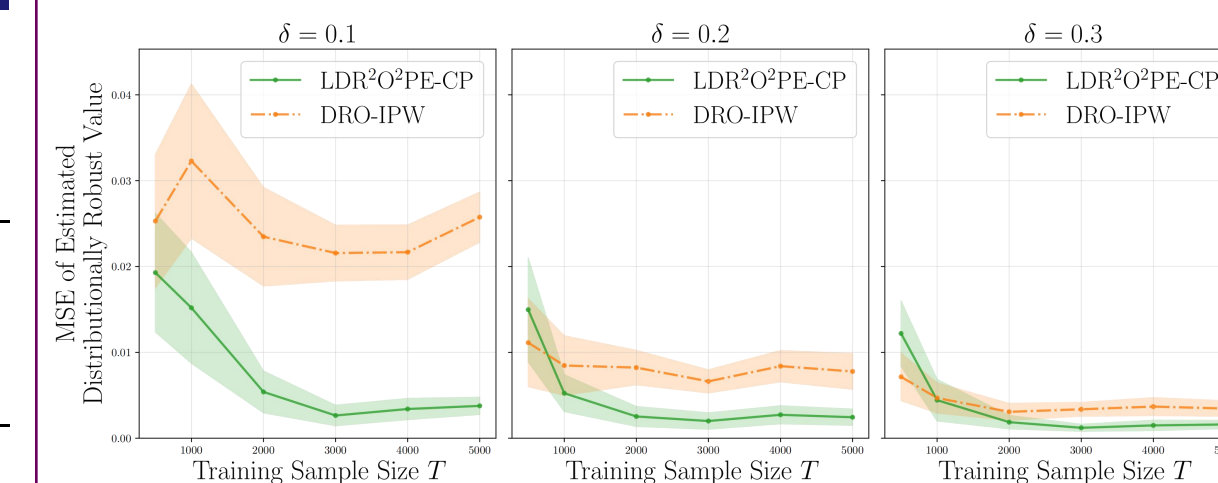
VALUATION: $f(X) = \theta_0 + \theta_1 X + \theta_2 X_1 X_2$

NOISE: $Z \sim \text{Logistic}(0, 0.2)$

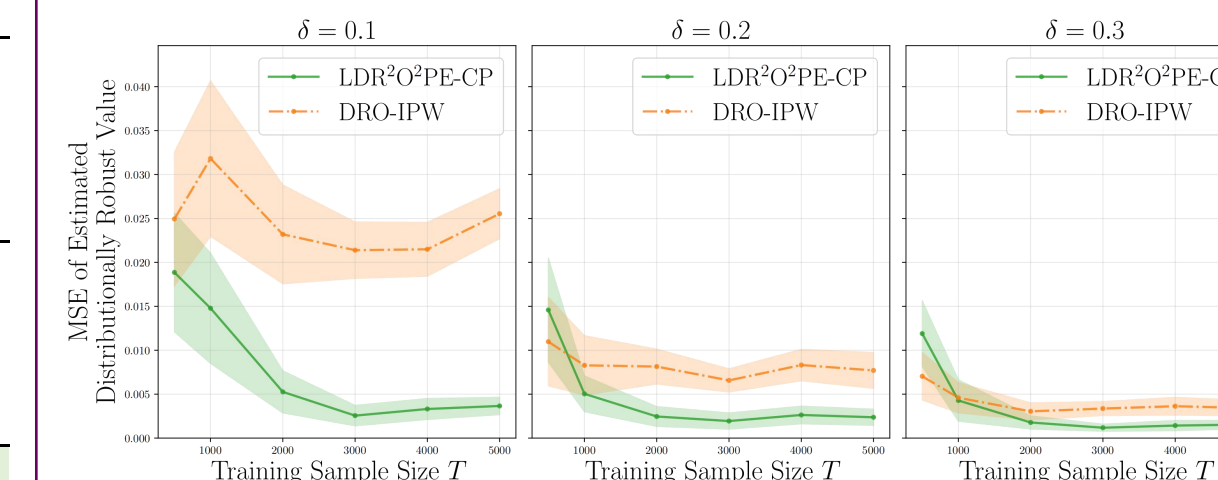
$\delta_{\text{train}} = 0.2$; Test $\delta \in \{0.1, 0.2, 0.3\}$

GRF (20 trees), $L=5$ folds, 20 seeds

OPE: MSE of R_δ vs. Sample Size T (lower=better)

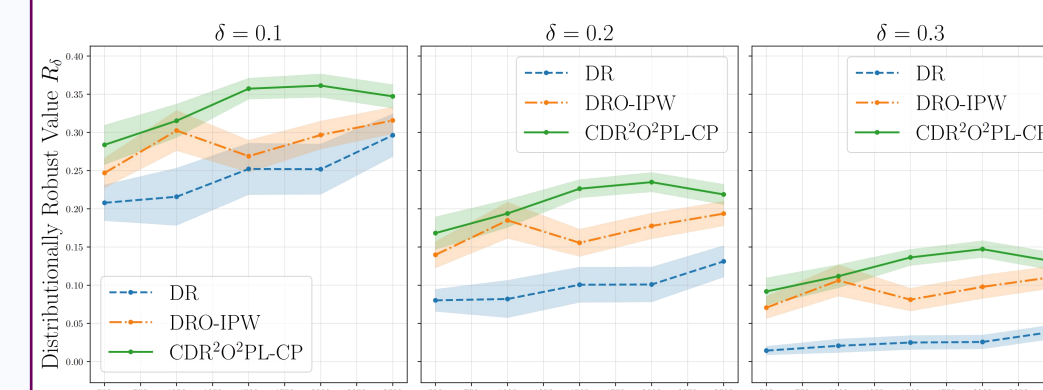


(a) $h = \Theta(T^{-1/4.9})$

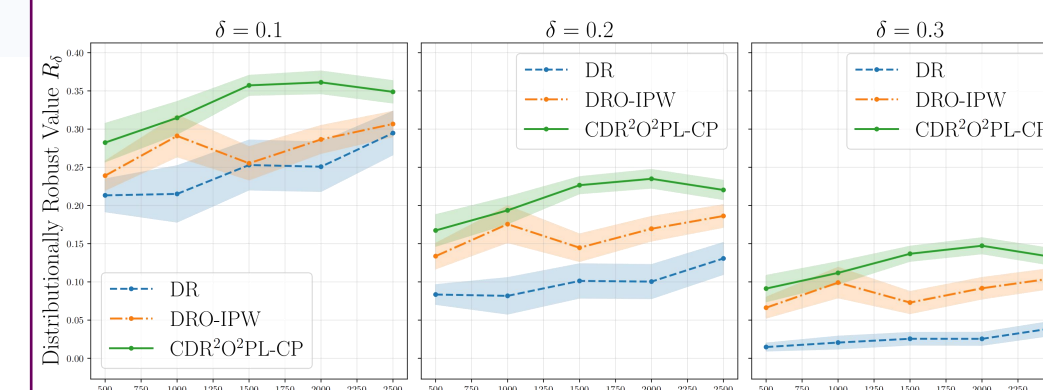


(b) $h = \Theta(T^{-1/5.0})$

OPL: Robust Value $R_\delta(\widehat{\pi})$ at $\delta=0.2$ (higher=better)



(a) $h = \Theta(T^{-1/4.9})$



(b) $h = \Theta(T^{-1/5.0})$

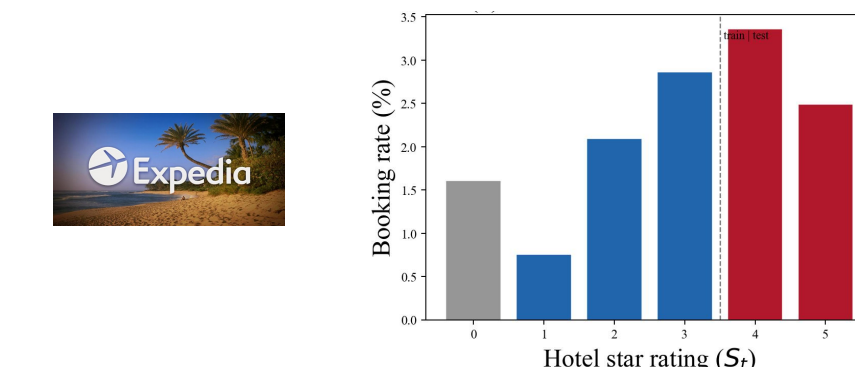


Figure: Outcome shift across hotel tiers

$$\hat{R}_{\text{adv}}(\pi) = \min_{1 \leq j \leq M} \frac{1}{T'} \sum_{t=1}^{T'} \widehat{r}(\widehat{x}_t^{(j)}, \pi(\widehat{x}_t^{(j)}))$$

$\widehat{r}(x, p)$ is a linear outcome regression for $\mathbb{E}[r|X = x, P = p]$

Table: Robust performance in Expedia study

Method	$\hat{R}_{\text{adv}}$
$\widehat{\pi}_{\text{Lin}}$	6.201 ± 0.683
$\widehat{\pi}_{\text{DRO-IPW}}$	6.254 ± 0.873
$\widehat{\pi}_{\text{CDR}^2\text{O}^2\text{PL-CP}}$	6.350 ± 0.598

✓ KEY FINDINGS

- OPE:** Faster MSE decay than DRO-IPW for all δ
- OPL:** Highest robust reward, tighter CI vs. DRO-IPW
- Non-robust DR highly sensitive to distributional shifts
- Both robust methods stable across $\delta \in \{0.1, 0.2, 0.3\}$

5. Resources

Paper



Code

